7 Radiative transfer: Absorption and emission

Reading for Day 7: Pierrehumbert Principles of Planetary Climate, Section 3.5-3.6.

We will now go beyond the blackbody approximation and consider absorption and emission of radiation through an idealised slab (i.e., patch of atmosphere) to derive the fundamental equation governing (non-scattering) radiative transfer.

7.1 Lambert's law and absorption

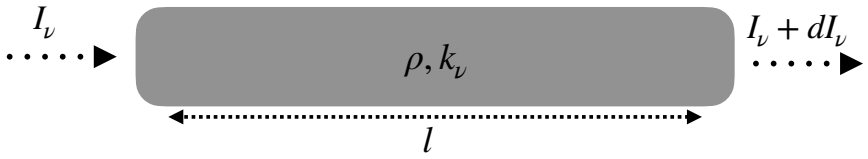


Figure 7.1: Purely absorbing slab with a density ρ, length l, and absorption coefficient k_ν (or k_λ).

Consider radiation with an initial radiance at a given wavelength I_λ (or frequency I_ν) that impinges upon an absorbing and/or emitting slab that has mass density ρ, absorption coefficient k_ν (units $\mathrm{m}^2\ \mathrm{kg}^{-1}$), and thermal radiance B_ν (in the case where it is emitting).

Let us first consider the case of a purely absorbing slab, with $B_\nu = 0$. Lambert's law is an empirical relationship which shows that the fractional change in radiance due to absorption is proportional to the mass per area in the slab. Thus, Lambert's law states that this slab absorbs radiation, causing a decrease in radiance leaving the slab (at distance dl) of

$$dI_\nu = -I_\nu k_\nu \rho dl \quad \text{Lambert's law.} \tag{7.1}$$

We can further define the *absorptivity* at any given frequency, which is the fraction of incident radiance of a given frequency absorbed by the material. In the case of an infinitesimal slab, $a_\nu = k_\nu \rho dl$, implying that we can express Lambert's law as

$$dI_\nu = -I_\nu a_\nu. \tag{7.2}$$

7.2 Kirchhoff's law and emission

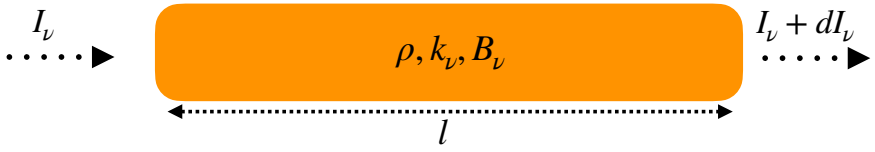


Figure 7.2: Emitting slab with a density ρ, length l, blackbody radiance B_ν, and (in the case of both an absorbing and emitting slab) an absorption coefficient k_ν (or k_λ).

In the last lecture, we introduced the *emissivity* ϵ_ν – this is the ratio of the radiance emitted by a substance to the blackbody radiance B_ν. Thus, blackbodies have $\epsilon_\nu = 1$, while real substances which are imperfect emitters rather than blackbodies have $\epsilon_\nu < 1$.

Kirchhoff's law states that substances in thermodynamic equilibrium emit as efficiently as they absorb. To briefly derive this, consider an energetically closed system where a slab with temperature T_1 is sandwiched between two slabs both with temperature T_0 with heat transfer between them accomplished purely via blackbody radiation. Thus, the energy balance of this middle slab implies that

$$2\bar{a}\sigma T_0^4 = 2\bar{\epsilon}\sigma T_1^4, \tag{7.3}$$

where $\bar{a}$ is the average absorptivity of the middle slab and $\bar{\epsilon}$ is the average emissivity of the middle slab. If radiative heat transfer between the slabs follows the second law of thermodynamics, this then implies that $T_0 = T_1$ and thus that $\bar{a} = \bar{\epsilon}$ – this holds equivalently at any given wavelength or frequency.

Thus,

$$\epsilon_\nu = a_\nu \quad \text{Kirchoff}'\text{s law.} \tag{7.4}$$

Note that Kirchhoff's law is only valid if the substance is in thermodynamic equilibrium. This is a consequence of the second law of thermodynamics (used in our derivation above) – for instance, if an object with $\epsilon \neq a$ is placed in a parcel of gas molecules emitting as blackbodies, conduction will attempt to equalise the temperatures, while radiation will try to make them different. This would lead to a violation of the second law of thermodynamics, as thermal energy would be transferred from the cold to the hot region – thus, $\epsilon = a$ (both at specific frequencies or wavelengths and integrated over all frequencies or wavelengths).

Therefore, the emission from an infinitesimal slab is $dI_\nu = \epsilon_\nu B_\nu$ which from Kirchhoff's law must be $dI_\nu = a_\nu B_\nu$. Replacing $a_\nu = k_\nu \rho dl$, the change in emitted radiation must be

$$dI_\nu = B_\nu k_\nu \rho dl \quad \text{emitting} - \text{only slab.} \tag{7.5}$$

7.3 Schwarzschild's equation

We can sum Lambert's and Kirchhoff's laws together to get the total change in intensity across the slab. Doing so and dividing through by dl, we can write Schwarzschild's equation for radiative transfer ignoring scattering:

$$\frac{dI_\nu}{dl} = \rho k_\nu (B_\nu - I_\nu) \quad \text{Schwarzschild}'\text{s equation.} \tag{7.6}$$

This equation tells us how the radiance from the slab is affected by a combination of absorption and emission, and we will provide solutions from it under certain idealisations in the following lecture.

For now, let us turn to discuss two related non-dimensional quantities (optical path and optical depth) which we will use throughout the radiative transfer module.

7.4 Optical path and optical depth

We can define the optical path $\tilde{\tau}$ as

$$\tilde{\tau}_\nu = \int_0^l \rho k_\nu dl', \tag{7.7}$$

which then implies $d\tilde{\tau}_\nu = \rho k_\nu dl$. This is a convenient measure of the amount of absorption/emission along a path followed by incident radiation, and as you will see it represents the e-folding distance for absorption/emission.

Our definition of τ allows us to re-write Schwarzschild's equation as

$$\frac{dI_\nu}{d\tilde{\tau_\nu}} = B_\nu - I_\nu. \tag{7.8}$$

We can define a related quantity to optical path, known as the optical depth τ, which is related to the optical path as

$$d\tau = -\mu d\tilde{\tau}, \tag{7.9}$$

where $\mu \equiv \cos\theta$. We can thus express Schwarzschild's equation ignoring scattering in optical depth coordinates as

$$\mu\frac{dI_\nu}{d\tau} = I_\nu - B_\nu. \tag{7.10}$$

Note that optical depth τ is essentially a vertical coordinate, and it increases downward (from the top of the atmosphere, as does pressure). This is because for upward-oriented radiances $\mu > 0$, and for downward-oriented radiances $\mu < 0$.

7.4.1 Photosphere pressure

It's clear from the above derivations that the optical depth (equivalent to the optical path if $\mu = 1$) is a critical parameter for estimating radiative properties of planetary atmospheres. If the optical depth $\tau \gg 1$, then the region is optically thick , implying $dI_\nu/d\tau \to 0$ and $I_\nu \approx B_\nu$. If the optical depth is $\tau \ll 1$, the region is optically thin, implying $dI_\nu/d\tau \ll 1$ and $I_\nu \approx I_\nu(0)$. However, if $\tau \approx 1$, then *both* incoming starlight is absorbed and radiation escapes to space – this special region of the atmosphere is called the "photosphere."

We can estimate the photosphere using the definition of optical depth,

$$\tau \approx 1 = \int_z^\infty k_\nu \rho dz = k_\nu \frac{p}{g}, \tag{7.11}$$

where we have integrated assuming hydrostatic balance ($-\rho dz = dp/g$).

If we assume that the absorption coefficient (equivalently, the opacity) is constant we can estimate the photosphere pressure where $\tau = 1$:

$$p_{\tau=1} \approx \frac{g}{k_\nu}. \tag{7.12}$$

For a typical hot Jupiter exoplanet with $g = 10$ m s^{-2} and $k_{\rm IR} = 4 \times 10^{-3}$ cm^2 g^{-1}, $p_{\rm IR} \approx 0.25$ bars. Jupiter itself has an absorption coefficient of $\sim 10^{-3}$ m^2 kg^{-1}, and a gravity of 26 m s^{-2}, implying $p_{\tau\approx1} \sim 0.3$ bars. Typically the thermal photosphere is at the ~ 100 mbar $-$ 1bar pressure level for irradiated planets.

7.5 Probing the thermal structure

The temperature-pressure profile can be directly probed by measuring spectral features in emission spectra of planets. Figure 7.3 demonstrates how the temperature-pressure profile is coupled to spectral features. In non-inverted atmospheres where the temperature decreases with increasing pressure, spectral features appear as dips in the emission spectrum at the wavelengths where the opacity from absorbers is large. This is because the radiatively active molecules absorb the thermal emission from below and re-emit it at their (cooler) local

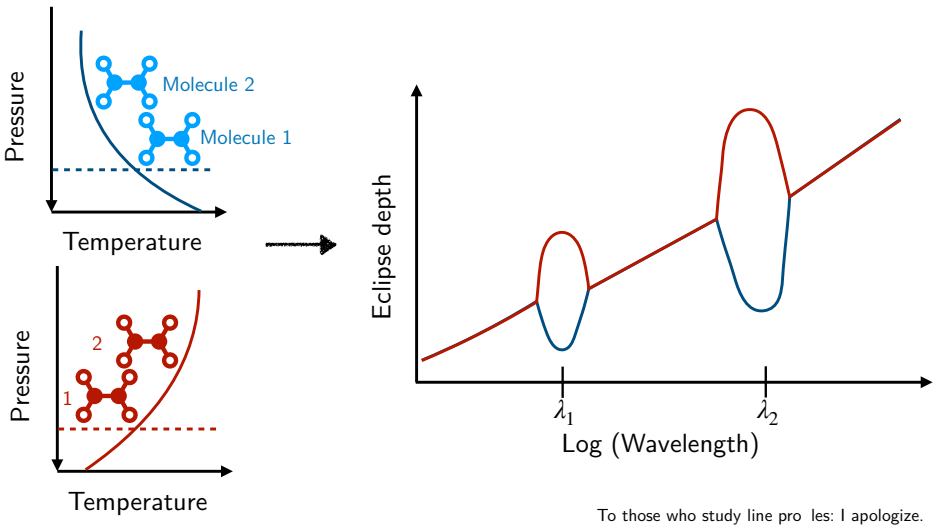


Figure 7.3: Cartoon demonstrating how absorption and emission features arise from non-inverted and inverted temperature pressure profiles, respectively. Note that here the y-axis of the spectrum shows the "eclipse depth" relevant for exoplanets, which is the planetary flux normalised by the flux of another blackbody (the star in this case).

temperature, causing a reduction in the planetary flux at that wavelength. In this case, molecule 2 has a larger relative dip (i.e., a larger absorption feature) because it is optically thick at lower pressures than molecule 1.

Conversely, in thermally inverted atmospheres there is a bump, or emission feature, rather than an absorption feature. This is because the molecules now are re-emitting the radiation they absorb at hotter local temperatures, causing an increase in the planet flux in the wavelengths at which they are opaque.

The fundamental ideas above can be generalised when knowing the full wavelength-dependence of the absorption coefficients for various gases to translate the observed thermal emission of a planet into an estimated temperature-pressure profile. Models which solve for this inverse problem are known as "retrievals," as they are benchmarked from Earth-observing satellites (as well as planetary missions) and now applied more broadly to telescopic observations of exoplanets.